

This page is the record for studying linear algebra, Hoffman & Kunze.

Assume  $T: V \rightarrow V$ ,  $V$  is finite dimensional vector space.

Lemma. Let  $\alpha \in V$  be an eigenvector,  $T\alpha = c\alpha$ .  
For any  $f \in F[x]$ ,  $f(T)\alpha = f(c)\alpha$ .

Thm. Let  $c_1, \dots, c_k \in F$  be distinct eigenvalues. Then,

$$\dim \sum_{i=1}^k E_{c_i} = \sum_{i=1}^k \dim E_{c_i}$$

Moreover, if  $\beta_i$  ( $1 \leq i \leq k$ ) is basis for  $E_{c_i}$ , then  $\bigcup_{i=1}^k \beta_i$  is basis for  $\sum_{i=1}^k E_{c_i}$   
Proof. Since  $W_1 + W_2 \leq V$  if  $W_1, W_2 \leq V$  and  $\langle \beta_1 \rangle + \langle \beta_2 \rangle = \langle \beta_1 \cup \beta_2 \rangle$

$$\sum_{i=1}^k E_{c_i} = \sum_{i=1}^k \text{span}(\beta_i) = \text{span}\left(\bigcup_{i=1}^k \beta_i\right)$$

Meanwhile, for each  $1 \leq i \leq k$ , let  $B_i \in E_{c_i}$ . Suppose that

$$B_1 + B_2 + \dots + B_k = 0.$$

For any polynomial  $f \in F[x]$ ,

$$0 = f(T)(0) = f(T)(B_1 + \dots + B_k) = f(T)(B_1) + \dots + f(T)(B_k) = f(c_1)B_1 + \dots + f(c_k)B_k$$

Using Lagrange's interpolation, choose  $f_i$  ( $1 \leq i \leq k$ ) s.t.

$$f_i(c_j) = \delta_{ij}.$$

Now,  $0 = \sum_{i=1}^k f_i(c_j) B_i = B_j$ , for each  $1 \leq j \leq k$ . In summary,

$$B_1 + \dots + B_k = 0 \quad (B_i \in E_{c_i}) \Rightarrow B_i = 0 \quad (1 \leq i \leq k)$$

Now, the set  $\bigcup_{i=1}^k \beta_i$  is linearly independent, because

if  $\sum_{i=1}^k \sum_{j=1}^{r_i} a_{i,j} \cdot B_{i,j} = 0$  ( $B_{i,j} \in \beta_i$ ), then  $\sum_{j=1}^{r_i} a_{i,j} B_{i,j} = 0$  ( $1 \leq i \leq k$ ),

thus all  $a_{i,j} = 0$ . Now Proved.

Thm. Let  $T: V \rightarrow V$  be a linear operator on a finite-dimensional space  $V$ .

Let  $c_1, \dots, c_k$  be the distinct eigenvalues of  $T$ . TFAE:

a).  $T$  is diagonalizable.

b).  $\text{ch}(T) = \prod_{i=1}^k (x - c_i)^{d_i}$  where  $d_i = \dim E_{c_i}$ .

c).  $\dim V = \sum_{i=1}^k \dim E_{c_i}$

## Minimal Polynomial.

Thm. The characteristic and minimal polynomials for  $T$  have the same roots.

Proof. Let  $p$  be the minimal polynomial for  $T$ .

Suppose that  $c \in F$  is root of  $p$ . That is,  $p(c) = 0$ . Then,

$$p = (x - c) \cdot q \quad (\deg q < \deg p)$$

for some  $q \in F[x]$ . Since  $p$  is the minimal,  $q(T) \neq 0$ . Take  $\beta \in V$  s.t.  $q(T)\beta \neq 0$ . Then,  $0 = p(T)(\beta) = (T - cI) \cdot q(T)(\beta)$ . Thus  $T \cdot q(T)(\beta) = c\beta$ ,  $c$  is eigenvalue.

Conversely, let  $c \in F$  be an eigenvalue of  $T$ . That is,  $T\beta = c\beta$  for some nonzero  $\beta \in V$ .

Now,  $0 = p(T)(\beta) = p(c)\beta$  and since  $\beta$  is nonzero,  $p(c) = 0$ .

Note; the Minimal polynomial is NOT necessarily irreducible; actually,

$$\mathcal{Q}_T: F[x] \rightarrow \mathcal{L}(V, V); f(x) \mapsto f(T)$$

the minimal polynomial is the generator of the ideal  $\ker \mathcal{Q}_T$ , where  $\mathcal{Q}_T$  is the ring-homomorphism. Since  $\mathcal{L}(V, V)$  is a ring under  $(+, \cdot)$  and it contains a zero divisor; for instance,  $\dim V = 2$ ,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0$$

Thus, even if  $p \in F[x]$  is the minimal polynomial,

$$0 = p(T) = f(T) \cdot g(T) \quad \text{for some } \deg f, \deg g < \deg p$$

can be happen.

Invariant.

(199.P)

Let  $T, U: V \rightarrow V$  be linear operators, and commute.

Both  $\text{Im } U$  and  $\text{Ker } U$  are invariant under  $T$ , because:

$$y \in \text{Im } U \Rightarrow y = U(x) \text{ for some } x \in V \Rightarrow T(y) = T(U(x)) = U(T(x)) \in \text{Im } U$$

$$y \in \text{Ker } U \Rightarrow U(y) = 0 \Rightarrow 0 = T(U(y)) = U(T(y)) \Rightarrow T(y) \in \text{Ker } U.$$

Since  $T$  commutes with  $g(T)$ , where  $g \in F[x]$ ,

$\text{Ker } g(T)$  and  $\text{Im } g(T)$  are invariant under  $T$ .

Lemma. Let  $W \leq V$  be an invariant subspace for  $T: V \rightarrow V$ . Then,

$\text{ch}(T_W)$  divides  $\text{ch}(T)$ . Moreover, the minimal polynomial of  $T_W$  divides minimal polynomial of  $T$ .

Proof. By the calculation,

$$[T]_{\beta} = \begin{bmatrix} [T_W]_{\beta'} & B \\ 0 & C \end{bmatrix} \quad \text{where } \beta \text{ and } \beta' \text{ are basis for } V \text{ and } W, \text{ respectively,} \\ \text{and } B' \subset \beta$$

$$\text{Now, } \text{ch}(T) = \det(xI - [T]_{\beta}) = \det(xI - [T_W]_{\beta'}) \cdot \det(xI - C) \\ = \text{ch}(T_W) \cdot \det(xI - C),$$

thus the first assertion is proved.

Conductor.

Def. Let  $W$  be an invariant subspace for  $T: V \rightarrow V$ , and  $\alpha \in V$  be a vector. Define  $T$ -Conductor of  $\alpha$  into  $W$  as;

$$S_T(\alpha; W) \stackrel{\text{def}}{=} \{g \in F[x] \mid g(T)(\alpha) \in W\}.$$

In particular, if  $W = \{0\}$ , the Conductor is called the  $T$ -annihilation of  $\alpha$ .

Lemma. The Conductor  $S(\alpha; W)$  is an Ideal in  $F[x]$ .

Proof. First, if  $T[W] \subseteq W$ , then  $g(T)[W] \subseteq W$ , because for any  $w \in W$ ,  
 $\beta \in W \Rightarrow T^n(\beta) = T^{n-1}(T(\beta)) = \dots = T(T^{n-1}(\beta)) \in W$ .

And since  $W \subseteq V$ , it is closed under the addition and scalar multiplication,  $g(T)[W] \subseteq W$ . Now, to show  $S(\alpha; W)$  is an Ideal, first show that it is a space. For any  $f, g \in S(\alpha; W)$  and  $c \in F$ ,  $(cf + g)(T) = cf(T) + g(T)$ , therefore

$$(cf + g)(T)(\alpha) = cf(T)(\alpha) + g(T)(\alpha) \in W$$

Thus it is a subspace. Moreover, for any  $f \in F[x]$  and any  $g \in S(\alpha; W)$ ,

$$(fg)(T)(\alpha) = f(T) \underbrace{g(T)(\alpha)}_{\in W} \in W.$$

Thus  $fg \in S(\alpha; W)$ .  $\square$

Note: Given the linear operator  $T$  and  $T$ -invariant subspace  $W$ , and any vector  $\alpha \in V$ ,

$$Q: F[x] \rightarrow V/W : f(x) \mapsto f(T)(\alpha) + W$$

is  $F[x]$ -module homomorphism, so  $S_T(\alpha; W) = \ker Q$ .

Lemma. If the minimal polynomial  $P(x) \in F[x]$  for  $T$  factors as:

$$P = (x - c_1)^{r_1} \dots (x - c_k)^{r_k}, \quad c_i \in F.$$

Condition

and let  $W$  be a proper subspace of  $V$  which is invariant under  $T$ , then there is a vector  $\alpha \in V$  s.t.

$$\alpha \notin W$$

results

2) For some eigenvalue  $c \in F$  of  $T$ ,  $(T - cI)(\alpha) \in W$ .

Proof. Let  $\beta \in V \setminus W$  be given. Let  $q \in F[x]$  be the  $T$ -annihilator of  $\beta$  into  $W$ .

That is,  $q(T)(\beta) \in W$ . Since  $(q(x) = S_T(\alpha; W))$  and  $P(T)(\alpha) = 0 \in W$ ,  $P \in S_T(\alpha; W)$  therefore  $q$  divides  $P$ . And, since  $\beta \notin W$ ,  $q$  must be not constant. Now,

$$q = (x - c_1)^{e_1} \dots (x - c_k)^{e_k}, \quad 0 \leq e_i \leq r_i, \text{ at least one } e_i \text{ is positive.}$$

Let  $j$  be a integer s.t.  $e_j$  is positive. Then,

$$q = (x - c_j)^{e_j} \cdot h \quad \text{for some } h \in F[x].$$

Since  $q$  is the generator of  $S_T(\alpha; W)$ ,  $h(T)\beta \notin W$ . But,

$$(T - c_j I)(h(T)\beta) = q(T)(\beta) \in W.$$

Thm. Let  $V$  be a finite dimensional vector space over  $F$ , and  $T: V \rightarrow V$  be a linear.

$T$  is triangulable if and only if minimal polynomial of  $T$  splits over  $F$ .

Proof. If  $T$  can be represented by triangular, then clear, since:

the determinant of triangle is product of diagonals. Conversely, suppose that

$$P(x) = (x - c_1)^{r_1} \dots (x - c_k)^{r_k}$$

is the minimal polynomial of  $T$ . Since  $P$  shares the roots with  $\chi(T)$ ,  $c_i$ 's are eigenvalues.

Choose  $\alpha_1 \in V$  s.t.  $\alpha_1 \notin \{0\}$  and  $(T - c_1 I)(\alpha_1) \in \{0\}$ , for some eigenvalue  $c \in F$ .

Choose  $\alpha_2 \in V$  s.t.  $\alpha_2 \notin \langle \alpha_1 \rangle$  and  $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$ , //

$$T(\alpha_2) = c_1 \alpha_1 + c_2 \alpha_2$$

Thm. Let  $V$  be a vector space over  $F$ .

For any non-zero linear functional  $f \in V^*$ ,  $\ker f$  is a hyperplane of  $V$ .

Conversely, every hyperplane in  $V$  is the kernel for some linear functional on  $V$ .

Proof. Let  $f \in V^*$  be a non-zero linear functional. Since  $f$  is non-zero, choose  $\alpha \in V \setminus \ker f$  (or  $f(\alpha) \neq 0$ ). We shall show that:  $V = \ker f + \langle \alpha \rangle$ .

Let  $\beta \in V$  be given. Put  $c = f(\beta)/f(\alpha)$ . (It can be defined because  $f(\alpha) \neq 0$ ).

Then,  $f(\beta - c\alpha) = f(\beta) - cf(\alpha) = 0$ , thus  $\beta - c\alpha \in \ker f$ , or  $\beta \in \ker f + \langle \alpha \rangle$ .

Conversely, let  $N \subseteq V$  be the hyperplane. Let  $\alpha \in V \setminus N$ .

Since  $N$  is the hyperplane,  $N + \langle \alpha \rangle = V$ . Therefore, for each  $\beta \in V$ ,

$$\beta = \gamma + c\alpha \quad \text{for some } \gamma \in N, c \in F.$$

This  $\gamma$  and  $c$  are uniquely determined, because  $N \cap \langle \alpha \rangle = \{0\}$ .

Define  $g: V \rightarrow F$  as  $g(\beta) = c$ . Then,  $\ker g = N$ .

$$\begin{pmatrix} a_1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & a_3 \end{pmatrix}^2 = \begin{pmatrix} a_1^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & a_3^2 \end{pmatrix}$$

**Lemma.** Let  $V$  be a fin. dim v.s over  $F$ , and let  $T: V \rightarrow V$  be a linear operator. Suppose that the minimal polynomial of  $T$  splits over  $F$  as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper  $T$ -invariant subspace  $W < V$ , there exists a vector  $\alpha \in V$  s.t.  $\alpha \notin W$  and for some  $c_i$ ,  $(T - c_i I)(\alpha) \in W$ .

**Proof.** Let  $\beta \in V \setminus W$  be given, and let  $q \in F[x]$  be a  $T$ -conductor of  $\beta$  into  $W$ . Then,  $q$  divides  $p$ . And, since  $\beta \notin W$ ,  $q$  is not constant. Therefore

$$q = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k} \quad (0 \leq e_i \leq r_i)$$

and at least one of  $e_i$  is positive. Pick  $i$  s.t.  $e_i > 0$ . Then,

$$q = (x - c_i) \cdot h \quad \text{where } h = q / (x - c_i) \in F[x].$$

Since  $q$  is  $T$ -conductor of  $\beta$ ,  $h(T)(\beta) \notin W$ , but  $(T - c_i I)(h(T)(\beta)) = q(T)(\beta) \in W$ .

**Thm.** Let  $V$  be a finite dimensional vector space over  $F$ , and  $T: V \rightarrow V$  be a linear.  $T$  is Triangularizable if and only if minimal polynomial of  $T$  splits over  $F$ .

**Proof.** If  $T$  is triangularizable with respects to a basis  $\beta$ , then the characteristic polynomial is

$$f(x) = \det([T]_{\beta} - xI) = \prod_{i=1}^n (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{th diagonal element of } [T]_{\beta}$$

thus, proved. Conversely, choose  $\alpha_1 \in V$  s.t.  $\alpha_1 \notin \{0\}$ ,  $(T - c_1 I)(\alpha_1) \in \{0\}$  for some eigenvalue  $c_1 \in F$  of  $T$ . Choose  $\alpha_2 \in V$  s.t.  $\alpha_2 \notin \langle \alpha_1 \rangle$ ,  $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$ , it is possible because  $\langle \alpha_1 \rangle$  is  $T$ -invariant. Again, choose  $\alpha_3 \in V$  s.t.  $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$ ,  $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$ . It is possible again,  $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$ .

Now, we obtain  $\{\alpha_1, \dots, \alpha_n\}$  s.t.

$$T(\alpha_1) = c_1 \alpha_1$$

$$T(\alpha_2) = a_{21} \alpha_1 + c_2 \alpha_2$$

⋮

$$T(\alpha_n) = a_{n1} \alpha_1 + a_{n2} \alpha_2 + \cdots + c_n \alpha_n$$

Thm. Let  $V$  be a fin. d'im. v.s over  $F$ , and  $T: V \rightarrow V$  be a linear operator.  
 $T$  is diagonalizable if and only if minimal polynomial of  $T$  splits and separable.

Proof. Suppose that  $T$  is diagonalizable. Let  $c_1, \dots, c_k$  be distinct eigenvalues.  
 Then, for any eigenvector  $d \in V$ , there is a  $c_i \in F$  s.t.  $(T - c_i I)(d) = 0$ .  
 Therefore, for  $p = (x - c_1) \dots (x - c_k)$ ,

$$p(T) = (T - c_1 I) \dots (T - c_k I)(d) = 0$$

because  $\{T - c_1 I, \dots, T - c_k I\}$  commute. Since  $T$  is diagonalizable,  
 for any vector of  $V$  can be represented by linear combination of eigenvectors,  
 $p(T) = 0$

Thus proved. Conversely, let  $W$  be a spanned subspace by all of eigenvectors of  $T$ .  
 Suppose that  $W \neq V$ . By the construction,  $W$  is  $T$ -invariant. Therefore, by the Lemma,  
 there is  $\alpha \in V \setminus W$  and eigenvalue  $c_i \in F$  s.t.  $\beta = (T - c_i I)(\alpha) \in W$ .  
 Since  $\beta \in W$ ,  $\beta = \beta_1 + \dots + \beta_k$  where  $T(\beta_i) = c_i \beta_i$ ,  $1 \leq i \leq k$ . Therefore, for any  $h \in F[x]$

$$h(T)(\beta) = \sum_{i=1}^k h(c_i) \cdot \beta_i \in W.$$

Put  $p = (x - c_i)g$ . Also,  $g - g(c_i) = (x - c_i)h$ . Then,

$$g(T)(\alpha) - g(c_i)(\alpha) = h(T)(T - c_i I)(\alpha) = h(T)(\beta) \in W$$

And,  $0 = p(T)(\alpha) = (T - c_i I)g(T)(\alpha) = g(T)(T - c_i I)(\alpha) \in W$ .

Therefore  $g(c_i)(\alpha) \in W$ , and since  $\alpha \notin W$ ,  $g(c_i) = 0$ . Contradicts to  $p = (x - c_i)^2 g$ .

**Lemma.** Let  $f, g: V \rightarrow F$  be linear functionals on  $V$ .

$g = c \cdot f$  for some  $c \in F$  if and only if  $\ker f \leq \ker g$ .

**Proof.** If  $f = 0$ , then trivial. Assume  $f \neq 0$ . Then,  $\ker f$  is the hyper-space of  $V$ .  
Let  $\alpha \in V$  s.t.  $f(\alpha) \neq 0$ . put  $c = g(\alpha)/f(\alpha)$ . Then,  $g - cf$  is a linear functional s.t.  
 $\ker f \leq \ker(g - cf)$  and  $\alpha \in \ker(g - cf)$ . Thus  $\ker(g - cf) = V$ , or  $g - cf = 0$ .

**Thm.** Let  $g, f_1, \dots, f_r$  be linear functionals on  $V$ . Then,

$g$  is linear combination of  $\{f_1, \dots, f_r\}$  iff  $\bigcap_{i=1}^r \ker f_i \leq \ker g$ .

**Proof.** If  $g = \sum_{i=1}^r c_i f_i$  for some  $c_i \in F$ , then for any  $\alpha \in \bigcap_{i=1}^r \ker f_i$ ,  $g(\alpha) = \sum_{i=1}^r c_i f_i(\alpha) = 0$ .  
Conversely, we shall use the Mathematical induction for  $r \in \mathbb{N}$ .

By the previous Lemma, if  $r=1$ , it is proved. Suppose that it is held for  $r=k-1$ .

Let  $f_1, \dots, f_k$  be linear functionals on  $V$  s.t.

$$\bigcap_{i=1}^k \ker f_i \leq \ker g.$$

Let  $g, f_1, \dots, f_{k-1}$  be restricted by  $\ker f_k$ . That is,

$$f_i|_{\ker f_k}: \ker f_k \rightarrow F; \alpha \mapsto f_i(\alpha). \quad (i=0, \dots, k-1, f_0 = g)$$

And, let  $\alpha \in \ker f_k$  s.t. for all  $1 \leq i \leq k-1$ ,  $f_i(\alpha) = 0$ .

That is,  $\alpha \in \bigcap_{i=1}^k \ker f_i$  and therefore  $g|_{\ker f_k}(\alpha) = 0$ , by the hypothesis.

By the assumption of the induction,

$$g|_{\ker f_k} = \sum_{i=1}^{k-1} c_i f_i|_{\ker f_k} \quad \text{for some } c_i \in F$$

Put  $h = g - \sum_{i=1}^{k-1} c_i f_i$ . Now, for any  $\alpha \in \ker f_k$ ,  $h(\alpha) = 0$ , or  $\ker f_k \leq \ker h$ .

The previous Lemma gives that  $h = c_k f_k$  for some  $c_k \in F$ . Now,  $g = \sum_{i=1}^k c_i f_i$ .

Proof.

$$\begin{array}{ccccc}
 V & \xrightarrow{\alpha_i \mapsto f_i} & V^* & \longrightarrow & V^{**} \\
 T \downarrow & & \uparrow T^t & & \downarrow T^{tt} \\
 W & \xrightarrow{\beta_i \mapsto g_i} & W^* & \longrightarrow & W^{**}
 \end{array}$$

$$\dim \ker f = \dim V - 1 \quad \text{where } f \in V^*.$$

$$\ker f := \{\alpha \in V \mid f(\alpha) = 0\}$$

$$S^0 := \{f \in V^* \mid f(\alpha) = 0 \quad \forall \alpha \in S\}.$$

$$\bigcap_{i=1}^r \ker g_i \subseteq W$$

$$\begin{aligned}
 W &= \ker f_{k+1} \cap \ker f_{k+2} \cap \dots \cap \ker f_n. \\
 &= F^{n-k}
 \end{aligned}$$

$$\begin{array}{c}
 A_{11}x_1 + \dots + A_{1n}x_n = 0 \\
 \vdots \\
 A_{m1}x_1 + \dots + A_{mn}x_n = 0
 \end{array}$$

$$W \subseteq \ker f \Leftrightarrow \bigcap \ker g_i \subseteq \ker f.$$

For each  $1 \leq i \leq m$ , define

$$f_i: F^n \rightarrow F; (x_i) \mapsto \sum_{j=1}^n A_{ij}x_j$$

$$\ker f_1 \cap \ker f_2 \cap \dots \cap \ker f_m$$

$W \subseteq V$ ,  $\{g_1, \dots, g_r\}$  is basis for  $W^0$ .

$$W = \bigcap_{i=1}^r \ker g_i$$

Proof.  $W \subseteq \bigcap \ker g_i$  is trivial, because  $W \subseteq \ker g_i$  for all  $g_i \in W^0$ .

Conversely, let  $\{f_1, \dots, f_m\} \subseteq V^*$  s.t.  $W = \ker f_1 \cap \dots \cap \ker f_m$ . Then, for each  $1 \leq i \leq m$ ,

$$\begin{aligned}
 W \subseteq \ker f_i &\Leftrightarrow f_i \in W^0 \Leftrightarrow f_i \text{ is linear combination of } \{g_1, \dots, g_r\} \\
 &\Leftrightarrow \bigcap_{i=1}^r \ker g_i \subseteq \ker f_i
 \end{aligned}$$

This proves.

Consider a system of linear equation;

$$\begin{array}{ccccccc} A_{11}x_1 & + & \dots & + & A_{1n}x_n & = & 0 \\ \vdots & & & & \vdots & & \vdots \\ A_{m1}x_1 & + & \dots & + & A_{mn}x_n & = & 0 \end{array}$$

Define  $f_i: F^n \rightarrow F: (x_j) \mapsto (A_{ij} \cdot x_j)$  for each  $1 \leq i \leq m$ .

We want to find a subspace:  $\ker f_1 \cap \dots \cap \ker f_m$ . (has dimension  $\leq n-m$ )

1). If  $m < n$ , it has a non-trivial solution.

2). If  $m = n$  and

Def. Let  $W_i \leq V$ .

If  $\alpha_1 + \dots + \alpha_k = 0$  ( $\alpha_i \in W_i$ )  $\Rightarrow \alpha_i = 0$ ,

then  $W_i$ 's are independent.

Lemma. Let  $W = \sum W_i$  TFAE

1)  $W_i$ 's are independent.

2) For each  $2 \leq j \leq k$ ,  $W_j \cap (W_1 + \dots + W_{j-1}) = \{0\}$ .

3) If  $\beta_i$  is basis for  $W_i$ , then  $\beta_1 \cup \dots \cup \beta_k$  is basis for  $W$ .

proof. 1)  $\Rightarrow$  2)

Let  $\alpha \in W_j \cap \sum_{i=1}^{j-1} W_i$ . Then,  $\alpha = \alpha_j = \alpha_1 + \dots + \alpha_{j-1}$ . Then,

$$\alpha_1 + \dots + \alpha_{j-1} - \alpha_j = 0 \Rightarrow \alpha_j = 0.$$

2)  $\Rightarrow$  1). Suppose  $\alpha_1 + \dots + \alpha_k = 0$  ( $\alpha_i \in W_i$ ). If not all zero, let  $j$  be the largest integer st  $\alpha_j \neq 0$ . Then,  $0 = \alpha_1 + \dots + \alpha_j \Rightarrow \alpha_j = -\alpha_1 - \dots - \alpha_{j-1} \in W_j \cap \{\sum_{i=1}^{j-1} W_i\}$ .

Def. A linear operator  $E: V \rightarrow V$  is called a projection if  $E^2 = E$ .

•  $x \in \text{Im } E \iff E(x) = x$

Proof.  $x \in \text{Im } E \Rightarrow x = E(\alpha)$  for some  $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$ . Converse is trivial.

•  $V = \text{Im } E \oplus \text{Ker } E$

Proof. Given  $x \in V$ ,  $x = E(x) + (x - E(x))$ . Clearly  $E(x) \in \text{Im } E$ , and

$$E(x - E(x)) = E(x) - E^2(x) = 0 \Rightarrow x - E(x) \in \text{Ker } E$$

And, let  $y \in \text{Im } E \cap \text{Ker } E$  be given. Then,  $E(y) = y$  and  $E(y) = 0$ , thus  $y = 0$ .

Thm. Let  $V = W_1 \oplus W_2$ . Then, there is one and only one projection  $E: V \rightarrow V$  s.t.

$$\text{Im } E = W_1 \quad \text{and} \quad \text{Ker } E = W_2$$

Proof. Ask a teacher on the street.

Let  $E$  be a projection. For  $\beta_1$ , basis for  $\text{Im } E$ , and  $\beta_2$ , basis for  $\text{Ker } E$ ,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \quad \text{where } r = \text{rank } E.$$

Thm. Let  $V$  be a fin. dim. v.s over  $F$ .

If  $V = \bigoplus_{i=1}^k W_i$ , then there exist  $k$  linear operators  $E_i$  on  $V$  s.t.

$$E_i \circ E_j = \delta_{ij} \cdot E_i, \quad I = \sum_{i=1}^k E_i, \quad W_i = \text{Im } E_i \quad (1 \leq i \leq k). \quad \dots (*)$$

Conversely, if  $E_1, \dots, E_k$  are  $k$  linear operators on  $V$  which satisfy  $(*)$ , then

$$V = \bigoplus_{i=1}^k W_i$$

Proof. Suppose  $V = \bigoplus_{i=1}^k W_i$ . For each  $1 \leq i \leq k$ , define  $E_i: V \rightarrow V: \sum_{j=1}^k \alpha_j \mapsto \alpha_i$ . proved.

Conversely, suppose that  $E_1, \dots, E_k$  are linear operators on  $V$  satisfying  $(*)$ .

Given  $\alpha \in V$ ,  $\alpha = E_1(\alpha) + \dots + E_k(\alpha) \in W_1 + \dots + W_k$ . And, to show uniqueness such expression,

let  $\alpha = \alpha_1 + \dots + \alpha_n$ , where  $\alpha_i \in W_i$ . And, let  $\alpha_i = E_i(\beta_i)$ , for each  $1 \leq i \leq k$ .

Now, for each  $i$ ,  $1 \leq i \leq k$ ,

$$E_j(\alpha) = \sum_{i=1}^k E_j(\alpha_i) = \sum_{i=1}^k (E_j \circ E_i)(\beta_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus Proved.

Thm. Let  $V$  be a fin. dim. Vs and  $T: V \rightarrow V$  be a linear operator.

Let  $E_1, \dots, E_k$  be projection on  $V$  s.t.  $E_i \circ E_j = \delta_{ij} \cdot E_i$ ,  $I = \sum_{i=1}^k E_i$ .

Then, each  $\text{Im } E_i$  be invariant under  $T$  if and only if  $T \circ E_i = E_i \circ T$ , for all  $1 \leq i \leq k$ .

Proof. Suppose that  $T$  commutes with each  $E_i$ . Then, for any  $\alpha \in \text{Im } E_i$ ,

$$T(\alpha) = T(E_i(\alpha)) = E_i(T(\alpha)) \in \text{Im } E_i.$$

Thus proved.

Conversely, suppose that each  $\text{Im } E_i$  is invariant under  $T$ . Given  $\alpha \in V$ , by the assumption,

$$\alpha = E_1(\alpha) + \dots + E_k(\alpha) \Rightarrow T(\alpha) = T(E_1(\alpha)) + \dots + T(E_k(\alpha))$$

And since each  $\text{Im } E_i$  is  $T$ -invariant,  $T(E_i(\alpha)) = E_i(\beta_i)$  for some  $\beta_i \in V$ . Now,

$$(E_j \circ T)(\alpha) = (E_j \circ E_i)(\beta_i) = \delta_{ij} \cdot E_i(\beta_i)$$

$$\text{Finally, } (E_j \circ T)(\alpha) = (E_j \circ T \circ E_1)(\alpha) + \dots + (E_j \circ T \circ E_k)(\alpha) = E_j(\beta_j) = (T \circ E_j)(\alpha). \quad \square$$

Thm. Let  $T$  be a linear operator on fin. dim  $V$ .

If  $T$  is diagonalizable with distinct eigenvalues  $C_1, \dots, C_k$ , then there are linear operators  $E_1, \dots, E_k$

$$T = C_1 E_1 + \dots + C_k E_k, \quad I = E_1 + \dots + E_k, \quad E_i \circ E_j = \delta_{ij} \cdot E_i, \quad \text{Im } E_i = \ker(T - C_i I).$$

Conversely, if there are linear operators  $E_i$  and distinct scalars  $C_i$  satisfying the condition above, then  $T$  is diagonalizable and has  $C_i$ 's as eigenvalues.

Proof. Suppose  $T$  is diagonalizable with distinct eigenvalues  $C_1, \dots, C_k \in F$ . Then,

$$V = \ker(T - C_1 I) \oplus \dots \oplus \ker(T - C_k I)$$

Define  $E_i$  be a projection onto  $\ker(T - C_i I)$ . Then, given  $\alpha \in V$ ,

$$\alpha = E_1(\alpha) + \dots + E_k(\alpha) \quad \text{and so}$$

$$T(\alpha) = T(E_1(\alpha)) + \dots + T(E_k(\alpha)) = C_1 E_1(\alpha) + \dots + C_k E_k(\alpha) = (C_1 E_1 + \dots + C_k E_k)(\alpha).$$

Conversely,

[Thm. Primary decomposition Theorem.

Let  $T$  be a linear operator on fin. dim v.s  $V$  over a field  $F$ .

Let  $p$  be the minimal polynomial for  $T$ , with factorization

$$p = p_1^{r_1} \cdots p_k^{r_k} \quad (p_i \text{ are irreducibles, } r_i \in \mathbb{N}).$$

Then, for  $W_i = \ker p_i(T)^{r_i}$  for each  $(1 \leq i \leq k)$ ,

1).  $V = W_1 \oplus \cdots \oplus W_k$

2). Each  $W_i$  is  $T$ -invariant.

3).  $T|_{W_i}$  has minimal polynomial  $p_i^{r_i}$ .

Proof. For each  $1 \leq i \leq k$ , define  $f_i = \prod_{j \neq i} p_j^{r_j}$ . Since  $p_1, \dots, p_k$  are distinct prime elements, so  $f_1, \dots, f_k$  are relatively prime. That is,

$$(f_1, \dots, f_k) = (f_1) + \cdots + (f_k) = 1. \quad \text{+ More specifically, } (f_1, f_2, \dots, f_m) = \left( \prod_{j=1}^m p_j^{r_j} \right)$$

Now, there are polynomials  $g_1, \dots, g_k$  s.t.

$$\sum_{i=1}^k f_i g_i = 1.$$

Moreover,  $f_i f_j$  is divided by  $p$  if  $i \neq j$ . Let  $E_i = (f_i g_i)(T)$ , for each  $i$ . Then,

$$E_1 + \cdots + E_k = I \quad \text{and} \quad E_i \circ E_j = 0 \quad \text{if } i \neq j,$$

the second assertion is proved by

$$E_i \circ E_j = (f_i g_i)(T) \circ (f_j g_j)(T) = (g_i g_j)(T) \circ (f_i f_j)(T) = 0 \quad \text{if } i \neq j.$$

Therefore, these  $E_1, \dots, E_k$  are projections. Moreover  $\text{Im } E_i = W_i = \ker p_i(T)^{r_i}$ , because let  $\alpha \in \text{Im } E_i$ . Then,  $\alpha = E_i(\alpha)$ , and so

$$p_i(T)^{r_i}(\alpha) = p_i(T)^{r_i} \circ E_i(\alpha) = p_i(T)^{r_i} \circ f_i(T) \circ g_i(T)(\alpha) = 0.$$

Conversely, let  $\alpha \in \ker p_i(T)^{r_i}$ . If  $j \neq i$ , then  $f_j g_j$  is divided by  $p_i^{r_i}$ , and so

$$f_j(T) \circ g_j(T)(\alpha) = 0.$$

And thus,  $E_i(\alpha) = \alpha$ ,  $\alpha \in \text{Im } E_i$ .

Inner product. Let  $F = \mathbb{R}$  or  $F = \mathbb{C}$ .

Def. Inner product.

Let  $V$  be a vector space over  $F$ . A map

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow F; (\alpha, \beta) \mapsto \langle \alpha, \beta \rangle$$

is called 'inner product' if it satisfies the following: for any  $\alpha, \beta, \gamma \in V$  and  $c \in F$ ,

1)  $\langle c\alpha + \beta, \gamma \rangle = c\langle \alpha, \gamma \rangle + \langle \beta, \gamma \rangle$

2)  $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$

3) If  $\alpha \neq 0$ , then  $\langle \alpha, \alpha \rangle > 0$

Lemma. An orthogonal set of non-zero vectors is linearly independent.

Proof. Let  $S \subseteq V \setminus \{0\}$  be an orthogonal set.

Suppose that  $\alpha_1, \dots, \alpha_n \in S$  are distinct vectors s.t.

$$\beta = c_1 \alpha_1 + \dots + c_n \alpha_n \text{ for some } c_i \in F.$$

Then, for each  $1 \leq k \leq n$ ,

$$\langle \beta, \alpha_k \rangle = \langle \sum c_i \alpha_i, \alpha_k \rangle = \sum c_i \langle \alpha_i, \alpha_k \rangle = c_k \langle \alpha_k, \alpha_k \rangle \Rightarrow c_k = \frac{\langle \beta, \alpha_k \rangle}{\langle \alpha_k, \alpha_k \rangle}$$

Since each  $\alpha_k \neq 0$ , if  $\beta = 0$ , then  $c_1 = \dots = c_n = 0$ .

Furthermore, if  $\beta \in V$  is linear combination of  $\alpha_1, \dots, \alpha_n \in S$ ,

$$\beta = \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i.$$

Gram-Schmidt process.

Let  $V$  be a fin. dim. inn. prod. space, let  $\{\beta_1, \dots, \beta_n\} \subseteq V$  be a linearly independent subset.

Then, there exists an orthogonal subset  $\{\alpha_1, \dots, \alpha_n\} \subseteq V$  s.t.

$$\text{Span}\{\alpha_1, \dots, \alpha_n\} = \text{Span}\{\beta_1, \dots, \beta_n\}.$$

Proof. Using mathematical induction for  $n$ . For  $n=1$ , take  $\alpha_1 = \beta_1$ , then clear.

Now, suppose that for  $m \in \mathbb{N}$ , the statement holds. Let  $\{\beta_1, \dots, \beta_{m+1}\}$  be lin. independent, and

$$\alpha_{m+1} = \beta_{m+1} - \sum_{i=1}^m \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i \text{ where } \alpha_1, \dots, \alpha_m \text{ are obtained by hypothesis.}$$

Then  $\alpha_{m+1} \neq 0$ , because  $\beta_{m+1}$  is independent with  $\{\alpha_1, \dots, \alpha_m\}$ . Moreover, for  $1 \leq j \leq m$ ,

$$\langle \alpha_{m+1}, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \sum_{i=1}^m \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \langle \alpha_i, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \langle \beta_{m+1}, \alpha_j \rangle = 0.$$

Now,  $\{\alpha_1, \dots, \alpha_{m+1}\}$  is orthogonal, non-zero set, thus linearly independent by the Lemma.

Moreover, it is subset of  $\text{Span}\{\beta_1, \dots, \beta_{m+1}\}$ , the replacement Theorem gives result.

Orthogonal Projection.

Thm. Let  $W$  be a subspace of an inner prod. space  $V$ , and let  $\beta \in V$ .

- 1). A vector  $\alpha \in W$  satisfies  $\forall \gamma \in W, \|\beta - \alpha\| \leq \|\beta - \gamma\|$  if and only if  $\beta - \alpha \in W^\perp$ .
- 2). If such vector in 1) exists, then it is unique.
- 3). If  $W$  is finite-dimensional with an orthonormal basis  $\{\alpha_1, \dots, \alpha_n\}$ , then

$$\alpha = \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i.$$

is the unique best approximation.

Proof. Suppose that for all  $\gamma \in W, \|\beta - \alpha\| \leq \|\beta - \gamma\|$ . Then,

$$0 \leq \|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 + 2 \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Take for any  $\gamma \in W, \gamma = \alpha - (\alpha - \gamma)$ , we can write

$$\|\gamma\|^2 + 2 \operatorname{Re} \langle \beta - \alpha, \gamma \rangle \geq 0 \quad \text{for all } \gamma \in W$$

Now, for any  $\mu \in W \setminus \{\alpha\}$ , take  $\gamma = \frac{-\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} \cdot (\alpha - \mu)$ . Then,

$$\frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} \geq 0$$

it holds iff  $\langle \beta - \alpha, \alpha - \mu \rangle = 0$ . Therefore  $\langle \beta - \alpha, \gamma \rangle = 0$  for all  $\gamma \in W$ .

Conversely, if  $\beta - \alpha \in W^\perp$ , then for any  $\gamma \in W$ ,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 \geq \|\beta - \alpha\|^2.$$

And, the uniqueness is given by: if  $\alpha_1, \alpha_2 \in W$  satisfies  $\beta - \alpha_1, \beta - \alpha_2 \in W^\perp$ , then

$$\text{for any } \gamma \in W, \langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0$$

$$\Rightarrow \langle \alpha_1 - \alpha_2, \gamma \rangle = 0$$

Since  $\alpha_1 - \alpha_2 \in W, \langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$ , thus  $\alpha_1 = \alpha_2$ .

And, the last assertion is proved by: for each  $1 \leq k \leq n$ ,

$$\langle \beta - \alpha, \alpha_k \rangle = \langle \beta, \alpha_k \rangle - \langle \alpha, \alpha_k \rangle = 0, \text{ therefore } \beta - \alpha \in W^\perp.$$

□

Def. If for all  $\beta \in V$ , there is such  $\alpha \in W$ , then

$$E: V \rightarrow V: \beta \mapsto \alpha$$

is orthogonal projection.

### Orthogonal Projection - Revised.

Let  $V$  be an inn. prod space, (not necessary finite dimension) and  $W \subseteq V$  be a subspace. For any vector  $\beta \in V$ , if  $\alpha \in W$  satisfies that:

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \dots (*)$$

Then, the  $\alpha$  is called best approximation to  $\beta$  by  $W$ , simply b.app.

Thm.  $\alpha \in W$  is b.app to  $\beta \in V$  by  $W$  iff for all  $\mu \in W$ ,  $\langle \beta - \alpha, \mu \rangle = 0$ .

Proof. For fixed  $\alpha \in W$  and  $\beta \in V$ , for any  $\gamma \in W$ ,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition (\*) holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now take  $\gamma = \alpha + \frac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} (\alpha - \mu)$  for any  $\mu \in W$  s.t.

$$\text{Then, } 0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = - \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff  $\langle \beta - \alpha, \alpha - \mu \rangle = 0$ , and all non zero vectors in  $W$  can be represented by  $\alpha - \mu$ , thus proved.

Conversely, if  $\beta - \alpha$  is orthogonal to every vector in  $W$ , then  $\operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle = 0$ , thus  $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$ .  $\square$

Thm. If b.app to  $\beta \in V$  by  $W$  exists, then it is unique.

Proof. Suppose that  $\alpha_1, \alpha_2 \in W$  satisfies for all  $\gamma \in W$ ,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take  $\gamma = \alpha_1 - \alpha_2$ , so  $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$ .  $\square$

Def. For all  $\beta \in V$ , there is b.app  $\alpha \in W$ , then the map

$$P: V \rightarrow V: \beta \mapsto \alpha$$

is called an orthogonal projection on  $W$ .

Thm. Let  $W$  be a fin. dim subspace of inn. prod space  $V$ .

Then, the orthogonal projection  $P: V \rightarrow V$  exists. Furthermore,

$P$  is idempotent linear operator with  $\text{Im } P = W$ ,  $\text{Ker } P = W^\perp$ .

Proof. Let  $\{\alpha_1, \dots, \alpha_n\}$  be an orthonormal basis for  $W$ . Then, a map

$$P: V \rightarrow V; \beta \mapsto \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$$

1). It maps  $\beta$  to orthogonal projection by  $W$ .

For each  $1 \leq i \leq n$ ,

$$\left\langle \beta - \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i, \alpha_i \right\rangle = \langle \beta, \alpha_i \rangle - \langle \beta, \alpha_i \rangle = 0$$

So,  $\beta - P(\beta) \in W^\perp$  for all  $\beta \in V$ . (And, such a vector is unique, thus well-defined)

2). It is linear.

Let  $\beta, \gamma \in V$  and  $c \in \mathbb{F}$  be given. Then,

$$P(c\beta + \gamma) = \sum \frac{\langle c\beta + \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i = c \cdot \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i + \sum \frac{\langle \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i = cP(\beta) + P(\gamma)$$

3). It is idempotent.

Given  $\beta \in V$ ,  $P(\beta) \in W$ . So,  $P^2(\beta) = P(\beta)$ .

4).  $\text{Im } P = W$ ,  $\text{Ker } P = W^\perp$ .

$\text{Im } P \subseteq W$  is clear, and for any  $\gamma \in W$ ,  $P(\gamma) = \gamma \in \text{Im } P$ . And,

$$\beta \in \text{Ker } P \Leftrightarrow P(\beta) = 0 \Leftrightarrow \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \leq i \leq n \Leftrightarrow \beta \in W^\perp.$$

Note: Let  $E: V \rightarrow V$  be a projection. In general,  $E$  can not be determined by its image.

For example, let  $V = \mathbb{R}^3$ ,  $W_1 = \text{Span}\{(1, 0, 0)\}$ ,  $W_2 = \text{Span}\{(0, 1, 0), (0, 0, 1)\}$ ,  $W_3 = \text{Span}\{(0, 1, 0), (1, 1, 1)\}$ .

Then,  $V = W_1 \oplus W_2 = W_1 \oplus W_3$ , but  $W_2 \neq W_3$ . Moreover,

$$E: V \rightarrow V; (x_1, x_2, x_3) \mapsto (x_1, 0, 0), \quad E': V \rightarrow V; (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$$

Both  $E$  and  $E'$  are projections with  $\text{Im } E = \text{Im } E' = \text{Span}\{(1, 0, 0)\}$ , but  $E \neq E'$ ,

$\text{Ker } E = W_2$  and  $\text{Ker } E' = W_3$ .

Corollary. Let  $\{\alpha_1, \dots, \alpha_n\}$  be an orthogonal set of non-zero vectors in inn. prod space  $V$ . Then,

$$\text{for all } \beta \in V, \quad \sum_{i=1}^n \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2, \text{ the equality holds iff } \beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i.$$

Proof. Take  $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|^2] \cdot \alpha_i$ . Then,  $\langle \beta - \gamma, \gamma \rangle = 0$ . So

$$\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \geq \|\gamma\|^2.$$

Riesz representation Theorem.

Thm. Let  $V$  be a fin. dim. inn. prod. v.s.

For any linear functional  $f: V \rightarrow F$ , there is a unique vector  $\beta \in V$  s.t.

$$\text{for all } \alpha \in V, \quad f(\alpha) = \langle \alpha, \beta \rangle$$

In other words,  $\mathcal{Q}: V \rightarrow V^*; \beta \mapsto \langle \cdot, \beta \rangle$  is a Conjugate isomorphism.

Proof. Let  $\{\alpha_1, \dots, \alpha_n\}$  be an orthonormal basis for  $V$ .

Given linear functional  $f: V \rightarrow F$ , put

$$\beta = \sum_{j=1}^n \overline{f(\alpha_j)} \cdot \alpha_j.$$

then, for each  $1 \leq k \leq n$ ,

$$\langle \alpha_k, \beta \rangle = \left\langle \alpha_k, \sum_{j=1}^n \overline{f(\alpha_j)} \alpha_j \right\rangle = \sum_{j=1}^n \overline{f(\alpha_j)} \cdot \langle \alpha_k, \alpha_j \rangle = \overline{f(\alpha_k)}$$

Clearly,  $\langle \cdot, \beta \rangle$  is linear, thus the existence is proved. And the uniqueness is shown:

if  $\langle \alpha, \beta \rangle = \langle \alpha, \gamma \rangle$  for all  $\alpha \in V$ , then for  $\alpha = \beta - \gamma$ ,  $\langle \beta - \gamma, \beta - \gamma \rangle = 0$ .

Thus,  $\beta = \gamma$ .  $\square$

Observe that a inn. prod. space  $V$  with orthonormal basis  $\{\alpha_1, \dots, \alpha_n\}$

Let  $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$  and  $\beta = y_1 \alpha_1 + \dots + y_n \alpha_n$ . Then,

$$\langle \alpha, \beta \rangle = x_1 \overline{y_1} + \dots + x_n \overline{y_n}$$

Meanwhile, for any linear functional  $f \in V^*$ ,

$$f(\alpha) = f(\alpha_1) x_1 + \dots + f(\alpha_n) x_n$$

If we want to  $\beta$  satisfies  $\langle \alpha, \beta \rangle = f(\alpha)$ , then take  $y_i = \overline{f(\alpha_i)}$ . Then is,

$$\beta = \overline{f(\alpha_1)} \alpha_1 + \dots + \overline{f(\alpha_n)} \alpha_n$$

Adjoint.

Def. Let  $T$  be a linear operator on inn. Prod. space  $V$ .

If there is a linear operator  $T^*$  s.t.

$$\langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle \text{ for all } \alpha, \beta \in V$$

Then such  $T^*$  is called adjoint of  $T$ .

Thm. If  $T$  is a linear operator on fin. dim inn. Prod space  $V$ ,  
then  $T$  has a unique adjoint map  $T^*$ .

Proof. By the Riesz representation Thm, given  $\beta \in V$ , there is a unique  $\beta' \in V$  s.t.  
for all  $\alpha \in V$ ,  $\langle T(\alpha), \beta \rangle = \langle \alpha, \beta' \rangle$ .

Define  $T^*: V \rightarrow V: \beta \mapsto \beta'$ . To show the linearity, let  $\beta, \gamma \in V$  and  $c, f$  be given.  
Then, for any  $\alpha \in V$ ,

$$\begin{aligned} \langle \alpha, T^*(c\beta + \gamma) \rangle &= \langle T(\alpha), c\beta + \gamma \rangle = c \langle T(\alpha), \beta \rangle + \langle T(\alpha), \gamma \rangle = c \langle \alpha, T^*(\beta) \rangle + \langle \alpha, T^*(\gamma) \rangle \\ &= \langle \alpha, cT^*(\beta) \rangle + \langle \alpha, T^*(\gamma) \rangle = \langle \alpha, cT^*(\beta) + T^*(\gamma) \rangle. \end{aligned}$$

$$\text{So } T^*(c\beta + \gamma) = cT^*(\beta) + T^*(\gamma).$$

Thm. Let  $V$  be a fin. dim. inn. Prod. space and let  $\mathcal{B} = \{d_1, \dots, d_n\}$  be an orthonormal basis.  
For any linear operator on  $V$ ,

$$[T]_{\mathcal{B}} = \left( \langle T(d_j), d_i \rangle \right)_{i,j}.$$

Proof. For each  $1 \leq i \leq n$ ,

$$T(d_i) = \sum_{j=1}^n \langle T(d_i), d_j \rangle \cdot d_j$$

Thus proved.

Adjoint vs Transpose.

$T: V \rightarrow V$ .

Adjoint:  $T^*: V \rightarrow V: \beta \mapsto$

Transpose:  $T^t: V^* \rightarrow V^*: \alpha \mapsto \alpha \circ T$

If  $V$  is fin. dim, then  $V^* = \{ \langle \cdot, \beta \rangle \mid \beta \in V \}$ , so

$$\alpha \circ T = \langle \cdot, \beta \rangle \circ T = \langle T(\cdot), \beta \rangle$$

$$\text{s.t. } \langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle \text{ for all } \alpha, \beta \in V$$

$\Gamma$  preserves Inner product.

Def Let  $V$  and  $W$  be Inn. prod spaces over  $F$ .

We say that a linear map  $T: V \rightarrow W$  preserves inner product if for all  $\alpha, \beta \in V$ ,

$$\langle T(\alpha), T(\beta) \rangle = \langle \alpha, \beta \rangle.$$

Lemma Let  $V$  and  $W$  be f.n.s. over  $F$  with  $\dim V = \dim W < \infty$ .

For linear map  $T: V \rightarrow W$ ,  $TF_A E_i$ :

- 1).  $T$  preserves inner product.
- 2).  $T$  is isomorphism preserving inner product.
- 3).  $T$  maps every orthonormal basis for  $V$  onto orthonormal basis for  $W$ .
- 4).  $T$  maps a orthonormal  $\parallel$

proof. 1)  $\Rightarrow$  2)  $\|T(\alpha)\|^2 = \|\alpha\|^2$ , so  $\ker T = \{0\}$  2)  $\Rightarrow$  1) is trivial.

2)  $\Rightarrow$  3) is also trivial, if  $\{\alpha_1, \dots, \alpha_n\}$  is orthonormal basis for  $V$ , then  $\{T(\alpha_1), \dots, T(\alpha_n)\}$  is  $T_{00}$ ,  $\langle T(\alpha_i), T(\alpha_j) \rangle = \langle \alpha_i, \alpha_j \rangle = \delta_{ij}$ .

3)  $\Rightarrow$  4)  $\nRightarrow$ , 4)  $\Rightarrow$  1) Let  $\{d_1, \dots, d_n\}$  be an orthonormal basis with

$\{\tau(\alpha_1), \dots, \tau(\alpha_n)\}$  also orthonormal basis for  $W$ . Then, for any  $\alpha, \beta \in V$ ,  
 $\alpha = \sum x_i d_i, \beta = \sum y_i d_i$  f.s.  $x_i, y_i \in F$ . Now,

$$\langle T(\alpha), T(\beta) \rangle = \langle \sum x_i \cdot T(\alpha_i), \sum y_j \cdot T(\beta_j) \rangle = \sum_i \sum_j x_i \cdot \bar{y}_j \cdot \delta_{ij} = \sum_i x_i \bar{y}_i = \langle \alpha, \beta \rangle.$$

Thm Let  $V$  and  $W$  be inn. prod space over  $F$ , and let  $T: V \rightarrow W$  be a linear map. Then,

$T$  preserves inner product if and only if  $T$  preserves Norm.

Proof. One direction is trivial. Suppose that  $T$  preserves Norm. By the polarization identity,

$$\langle T(\alpha), T(\beta) \rangle = \frac{1}{4} \cdot \sum_{n=1}^4 i^n \cdot \|T(\alpha) + i^n \cdot T(\beta)\|^2 = \frac{1}{4} \cdot \sum_{n=1}^4 i^n \cdot \|T(\alpha + i^n \beta)\|^2 = \frac{1}{4} \cdot \sum_{n=1}^4 i^n \cdot \|\alpha + i^n \beta\|^2 = \langle \alpha, \beta \rangle$$

## Unitary Operator.

Def. Let  $V$  be an Inn. prod space.

An automorphism preserving inn. prod is called unitary operator.

Equivalently,  $U$  is unitary iff adjoint  $U^*$  exists and  $U^* \circ U = U \circ U^* = I$ .

Proof of the equivalence: Suppose that  $U$  is unitary. Then, for any  $\alpha, \beta \in V$ ,

$$\langle U(\alpha), \beta \rangle = \langle U(\alpha), (U \circ U^{-1})(\beta) \rangle = \langle \alpha, U^{-1}(\beta) \rangle.$$

So  $U^{-1} = U^*$ . Conversely, suppose  $U^*$  exists and  $U^* = U^{-1}$ . Then invertible is clear, and

$$\langle U(\alpha), U(\beta) \rangle = \langle \alpha, (U^* \circ U)(\beta) \rangle = \langle \alpha, \beta \rangle. \quad \square$$

Def. A real or complex Matrix  $A \in \text{Mat}_n$  is called orthogonal if  $A^t A = I$ .

Thm. For any  $B \in GL_n(\mathbb{C})$ , there exists a unique lower-triangular  $M$  with positive diagonal s.t.  $MB$  is unitary.

Proof. Since  $B$  is invertible, the row vectors  $\{\beta_1, \dots, \beta_n\}$  is basis for  $\mathbb{C}^n$ .

Let  $\{\alpha_1, \dots, \alpha_n\}$  be an orthonormal basis for  $\mathbb{C}^n$  obtained from  $\beta_i$ 's by Gram-Schmidt.

Now, for each  $1 \leq k \leq n$ ,  $\alpha_k = \beta_k - \sum_{i < k} \frac{\langle \beta_k, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$ . And since  $\text{span}\{\beta_1, \dots, \beta_{k-1}\} = \text{span}\{\alpha_1, \dots, \alpha_{k-1}\}$ ,

$$\alpha_k = \beta_k - \sum_{i < k} \overset{\text{unique}}{C_{ki}} \beta_i \quad \text{for some } C_{ki} \in \mathbb{C}.$$

$$\text{Let } U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \dots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix} \quad \text{and} \quad M_{ki} = \begin{cases} -\frac{1}{\|\alpha_k\|} \cdot C_{ki} & \text{if } i < k \\ \frac{1}{\|\alpha_k\|} & \text{if } i = k \\ 0 & \text{otherwise.} \end{cases}$$

↑ Revised in After page.

Unique decomposition Theorem. (OR QR decomposition)

For any  $B \in GL_n(\mathbb{C})$ , there exists a unique lower-triangular  $M$  with positive diagonal s.t  $MB$  is unitary,

Proof. Since  $B$  is invertible, the row vectors of  $B$ ,  $\{\beta_1, \dots, \beta_n\}$  is basis for  $V$ .

Apply Gram-Schmidt process, that is, for each  $1 \leq k \leq n$ , put

$$\alpha_k = \beta_k - \sum_{j < k} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \alpha_j$$

Then,  $\text{span}\{\alpha_1, \dots, \alpha_r\} = \text{span}\{\beta_1, \dots, \beta_r\}$  by induction, and  $\{\alpha_1, \dots, \alpha_n\}$  is orthonormal.

Put  $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \dots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$ , then clearly  $U^* U = I$ , because  $\{\alpha_1, \dots, \alpha_n\}$  is orthonormal

And, for each  $1 \leq k \leq n$ ,  $\alpha_k = \beta_k - \sum_{j < k} C_{kj} \beta_j$  for some  $C_{kj}$ .

Put  $M_{kj} = \begin{cases} -\frac{C_{kj}}{\|\alpha_k\|} & \text{if } k > j \\ \frac{1}{\|\alpha_k\|} & \text{if } k = j \\ 0 & \text{otherwise.} \end{cases}$  Then, each  $1 \leq k \leq n$ ,  $U_k = \begin{pmatrix} & & 0 \\ & & \\ M_{k1} & \dots & M_{kn} \end{pmatrix} \begin{pmatrix} \beta_1 & \dots & \beta_n \end{pmatrix} = U$

So  $MB = U$ . To show the uniqueness, denote

$$T^+(n) = \{M \in M_{nn}(\mathbb{C}) \mid M \text{ is lower triangular, positive diagonals}\}.$$

Let  $M_1, M_2 \in T^+(n)$  satisfies  $M_1 B, M_2 B$  are unitary. Since unitary matrices forms a group

$$(M_1 B) \cdot (M_2 B)^{-1} = M_1 M_2^{-1} \text{ is also unitary.}$$

And,  $M_2^{-1}$  is lower-triangular with positive diagonals, so is  $M_1 M_2^{-1}$ . Now,

$$(M_1 M_2^{-1})^{-1} = (M_1 M_2^{-1})^*, \text{ so } M_1 M_2^{-1} \text{ is diagonal.}$$

So,  $M_1 M_2^{-1} = I$ , or  $M_1 = M_2$ . @

Corollary. For each  $B \in GL(n)$ , there exists unique matrices  $K, A, N$  s.t  $B = KAN$ ,

where  $K$  is unitary,  $A$  is diagonal with positive values,  $N$  is upper-triangular with diagonals 1

+ Note. Let  $T_n(F)$  be the set of  $n \times n$  upper-triangular matrices with all diagonals are 1.

Then,  $(T_n(F), \cdot)$  is group, because: for any  $A \in T_n(F)$ ,  $A = I + N$ ,

where  $N$  is strictly upper triangular with  $N^n = 0$ , nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \dots + (-N)^{n-1},$$

and  $A \in T_n(F)$  is clear.

### QR - Factorization.

Let  $A = (w_1 \dots w_n)$  be an invertible matrix.

And, for each  $1 \leq k \leq n$ , 
$$V_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, V_i \rangle}{\|V_i\|^2} \cdot V_i.$$

Then,  $\{V_1, \dots, V_n\}$  is an orthogonal, so put  $U_k = V_k / \|V_k\|$ , then

$\{U_1, \dots, U_n\}$  is an orthonormal. Clearly,  $Q \stackrel{\text{def}}{=} (U_1 \dots U_n)$  is unitary.

Moreover, for each  $1 \leq k \leq n$ ,

$$w_k = \|V_k\| \cdot U_k + \sum_{i=1}^{k-1} \langle w_k, U_i \rangle \cdot U_i.$$

Define

$$R_{jk} = \begin{cases} \|V_k\| & j=k \\ \langle w_k, U_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then,  $R$  is upper-triangular, and satisfying

$$Q^{-1}A = Q^*A = \begin{pmatrix} \overline{U_1} \\ \vdots \\ \overline{U_n} \end{pmatrix} \cdot (w_1 \dots w_n) = \begin{pmatrix} \langle w_1, U_1 \rangle & \langle w_2, U_1 \rangle & \dots & \langle w_n, U_1 \rangle \\ \langle w_1, U_2 \rangle & - & - & \vdots \\ \vdots & - & - & \vdots \\ \langle w_1, U_n \rangle & - & - & \langle w_n, U_n \rangle \end{pmatrix} \\ = R.$$

so,  $A = QR$ .

Matrix Decomposition.

In this paper, our ambient space is  $M_{n \times n}(F)$ , simply  $M$ .

Given  $A \in M$ , denote  $A^{[i]}$  is  $i$ 'th row of  $A$ ,  $A_{[j]}$  is  $j$ 'th column of  $A$ .

With standard inner product on  $F^n$ , we can write

$$A \cdot B = \left( \langle A^{[i]}, B^{[j]} \rangle \right)_{i,j}.$$

$$\text{and } \langle A^{[i]}, B^{[j]} \rangle = \sum_{k=1}^n A_{ki} \cdot B_{kj}$$

Cyclic subspace.

In this, every vector space is finite-dimensional, and  $T: V \rightarrow V$  is arbitrary linear operator.  
Def. Given a vector  $\alpha \in V$ , define  $T$ -cyclic subspace generated by  $\alpha$  is:

$$Z(\alpha; T) \stackrel{\text{def}}{=} \{g(T)(\alpha) \mid g(t) \in F[t]\} = \text{span}_F \{T^k(\alpha) \mid k \geq 0\}$$

If  $Z(\alpha; T) = V$ , then  $\alpha$  is called a cyclic vector for  $T$ .

Def. Given  $\alpha \in V$ , the kernel of a map

$$\varphi_T^\alpha: F[t] \rightarrow V; g(t) \mapsto g(T)(\alpha)$$

is called  $T$ -annihilator of  $\alpha$ , and since  $F[t]$  is P.I.D., it has unique monic generator.

Denote  $\ker \varphi_T^\alpha = M(\alpha; T) = (P_\alpha)$  where  $P_\alpha$  is monic.

And,  $P_\alpha$  divides the minimal polynomial of  $T$ .

Thm. Given non-zero vector  $\alpha \in V$ , and  $T$ -annihilator of  $\alpha$ , the followings hold:

- 1).  $\dim Z(\alpha; T) = \deg P_\alpha$ .
- 2).  $Z(\alpha; T) = \text{span}_F \{T^i(\alpha) \mid 0 \leq i \leq k-1\}$  where  $k = \deg P_\alpha$ .
- 3). The restriction  $T|_{Z(\alpha; T)}$  has a minimal polynomial as  $P_\alpha$ .

Proof. Let  $g(t) \in F[t]$  be given. Then, by the division algorithm,

$$g = P_\alpha \cdot \bar{g} + r \quad (r=0 \text{ or } \deg r < \deg P_\alpha)$$

So,  $g(T)(\alpha) = (P_\alpha \cdot \bar{g})(T)(\alpha) + r(T)(\alpha) = r(T)(\alpha) \in \text{span}_F \{T^i(\alpha) \mid 0 \leq i \leq k-1\}$ .

And, these are linearly independent, by the minimality of  $P_\alpha$ .

To show the last assertion, check that  $Z(\alpha; T)$  is  $T$ -invariant. Now, given  $g \in F[t]$ ,

$$(P_\alpha(T|_Z) \circ g(T))(\alpha) = (P_\alpha(T) \circ g(T))(\alpha) = g(T)(\alpha) = 0.$$

And, by the minimality of  $P_\alpha$ , it is also minimal polynomial of  $T|_Z$ .

# The Companion Matrix.

Let  $U$  be a linear operator on  $W$  of dimension  $k \in \mathbb{N}$ , which has a cyclic vector  $\alpha$ .

Let  $\alpha_i = U^{i-1}(\alpha)$ ,  $i=1, \dots, k$ . Then,  $\{\alpha_1, \dots, \alpha_k\}$  is ordered basis for  $U$ , and

$\alpha_{i+1} = U^i(\alpha) = U(\alpha_i)$ ,  $i=1, \dots, k-1$ , and for minimal polynomial  $p_\alpha$  of  $U$ ,

$$p_\alpha(U)(\alpha) = U^k(\alpha) + c_{k-1}U^{k-1}(\alpha) + \dots + c_1U(\alpha) + c_0\alpha = 0.$$

So  $U^k(\alpha) = U(\alpha_k) = -c_0\alpha_1 - c_1\alpha_2 - \dots - c_{k-1}\alpha_k$ . Now, for  $\mathcal{B}$

$$[U]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & \dots & 0 & -c_0 \\ 1 & 0 & & 0 & -c_1 \\ 0 & 1 & & 0 & \vdots \\ \vdots & 0 & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -c_{k-1} \end{bmatrix} \text{ is called the Companion Matrix of } p_\alpha.$$

Def. Let  $V$  be an f.n.d.m. inn. prod. space.

A linear operator  $T: V \rightarrow V$  is called Normal if  $T \circ T^* = T^* \circ T$ .

+ Notes the existence of adjoint is guaranteed by the Riesz Representation Theorem.

Thm. Let  $V$  be an f.n.d.m. inn. prod. space, and  $T$  be a Hermitian.

Then, every eigenvalue of  $T$  is real, and eigenvectors of distinct eigenvalues are orthogonal.

Proof. Given eigenvalue  $c$  of  $T$ , for any eigenvector  $\alpha \in V$  corresponding to  $c$ , observe

$$c \cdot \langle \alpha, \alpha \rangle = \langle c\alpha, \alpha \rangle = \langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \langle \alpha, c\alpha \rangle = \overline{c} \cdot \langle \alpha, \alpha \rangle$$

$\xrightarrow{\text{Def}} \quad \xrightarrow{T\alpha=c\alpha} \quad \xrightarrow{V.f.n. \text{ adj. exist}} \quad \xrightarrow{\text{Hermitian}} \quad \xrightarrow{T\alpha=c\alpha} \quad \xrightarrow{\text{Def}}$

Since  $\langle \alpha, \alpha \rangle$  is positive real number ( $\because \alpha \neq 0$ ), so  $c = \overline{c}$ , or  $c \in \mathbb{R}$ .

Moreover, let  $c_1, c_2 \in \mathbb{R}$  be distinct eigenvalues, and  $\alpha_1, \alpha_2$  be eigenvectors of  $c_1, c_2$ , respectively. Then,

$$c_1 \cdot \langle \alpha_1, \alpha_2 \rangle = \langle c\alpha_1, \alpha_2 \rangle = \langle T(\alpha_1), \alpha_2 \rangle = \langle \alpha_1, T^*(\alpha_2) \rangle = \langle \alpha_1, T(\alpha_2) \rangle = \langle \alpha_1, c_2 \alpha_2 \rangle = \overline{c_2} \langle \alpha_1, \alpha_2 \rangle$$

By the first assertion,  $\overline{c_2} = c_2$ , so  $(c_1 - c_2) \cdot \langle \alpha_1, \alpha_2 \rangle = 0$ .

Finally,  $\mathbb{C}$  has no zero divisor,  $\langle \alpha_1, \alpha_2 \rangle = 0$ . (A kind of take.)

Thm. Every Hermitian on f.n.d.m. inn. prod. space has an eigenvalue.

Proof. Set  $V$  be a such space, and  $T$  is Hermitian. Take  $\mathcal{B}$  be an orthonormal basis for  $V$ .

Let  $A = [T]_{\mathcal{B}}$ . Then,  $A = [T^*]_{\mathcal{B}} = [T]_{\mathcal{B}}^* = A^*$ .

Put  $W \stackrel{\text{def}}{=} \{X \in M_{n \times 1}(\mathbb{C})\}$ , with inner product  $\langle X, Y \rangle = Y^* X$ .

Then, a map  $H: W \rightarrow W: X \mapsto AX$  is Hermitian. Moreover, the characteristic polynomial of  $H$ .

$$\det(xI - H) = \det(xI - A)$$

is a polynomial of degree  $n$  over  $\mathbb{C}$ , so it has a root  $c \in \mathbb{C}$ .

Thus,  $c$  is an eigenvalue of  $H$ , Hermitian, so it is a real number.

Note: Characteristic Polynomial of Hermitian splits over the given field  $\mathbb{R}$  or  $\mathbb{C}$ .

**Lemma.** Let  $V$  be an fin. dim inn prod space, and  $T$  be a linear operator.

For any  $T$ -invariant subspace  $W \subset V$ , the orthogonal complement  $W^\perp$  is  $T^*$ -invariant.

**Proof.**  $W^\perp$  is  $T^*$ -invariant iff for any  $\alpha \in W^\perp$ ,  $T(\alpha) \in W^\perp$ .

Given  $\alpha \in W^\perp$ ,  $\langle \alpha, T(\beta) \rangle = 0$  for all  $\beta \in W$ , since  $T(\beta) \in W$ .

That is,  $\langle T^*(\alpha), \beta \rangle = 0$  for all  $\beta \in W$ , or  $T^*(\alpha) \in W^\perp$ .

**Thm.** Let  $V$  be an fin. dim inn prod space, and let  $T: V \rightarrow V$  be a linear operator.

$T$  is Hermitian if and only if There is an orthonormal basis for  $V$  consisted by eigenvectors of  $T$ .

**Proof.** Using induction for  $\dim V = n$ .

Since  $T$  is Hermitian, it has at least one eigenvector  $\alpha_1 \in V$ . Now,

$\{\alpha_1 / \|\alpha_1\|\}$  is the orthonormal basis we desired. Now, let the statement hold for  $n-1$ .

Let  $\dim V = n$ . Then